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FOUNDATION**

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How to Do a Winning Science Fair Project

A step-by-step guide for students, from idea to presentation

Many schools lack the resources, knowledge, or connections to take part in science fairs. This guide exists to change that, so every student can compete on fair ground.

STUDENT WORKBOOK

Welcome, Future Scientist

A science fair project is a real piece of scientific work that you plan and run yourself, then explain to others. This guide walks you through every stage, from the first spark of an idea to the moment you face the judges.

HOW TO USE THIS WORKBOOK

Read each step, then use the write-in spaces, checklists, and worksheets to plan your own project. Work through it in order, or jump to the step you need. Keep it with your lab notebook so your thinking stays in one place.

What is inside

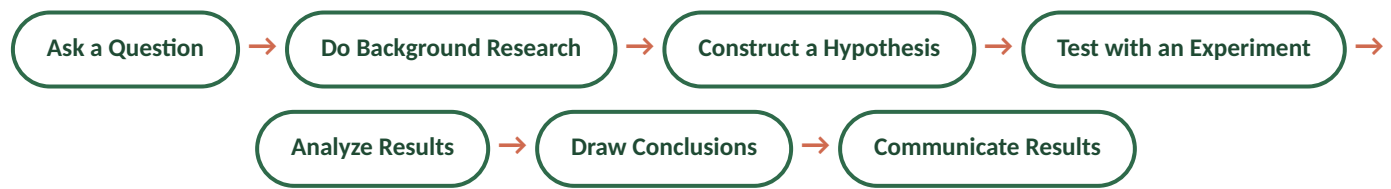
- **The Scientific Method:** the cycle every project follows
- **Steps 1 to 11:** from choosing a topic to presenting to judges
- **Project Timeline Checklist:** a month-by-month plan
- **Companion Resources:** templates and video tutorials that pair with this guide

A NOTE ON SAFETY AND HONESTY

Good science is safe science, and good scientists tell the truth about their results. Always work with adult supervision when a step calls for it, and record what really happened, even when the outcome surprises you.

The Scientific Method

Every strong project follows the same path. Scientists call it the scientific method. It is a cycle, not a straight line, so your results often send you back to ask a sharper question.



KEEP THIS CYCLE IN MIND

Each of the eleven steps in this guide fits somewhere on this cycle. When you feel stuck, find where you are on the path and look at what comes next.

WORKSHEET

IN ONE SENTENCE, WHAT QUESTION DO YOU WANT TO ANSWER?

1

Choosing Your Topic

Find a question that is interesting, doable, and safe

The best topic sits where your interest meets what you can actually complete. A project you care about keeps you motivated for months. A project you can finish on time and on budget keeps you sane.

The interest and feasibility test

Before you commit, ask yourself these questions:

- ☐ What am I curious about?
- ☐ Can I complete this in the time I have?
- ☐ Can I afford the materials?
- ☐ Is it safe to do?
- ☐ Can I find enough published research on this topic?

Research project or engineering project?

Type	What you do
Research	Investigate a question by running an experiment and analyzing data.
Engineering	Solve a problem by designing, building, and testing a solution.

WORKSHEET • PICK YOUR DIRECTION

- ☐ **Research project.** My question is: _____
- ☐ **Engineering project.** The problem I want to solve is: _____

Topic ideas by interest

If you love...	Explore...
Animals	Animal behavior, ecology, veterinary science
Sports	Sports science, biomechanics, nutrition
The environment	Climate, pollution, conservation, renewable energy
Technology	Coding, robotics, artificial intelligence, cybersecurity
Medicine	Disease research, public health, drug effectiveness
Space	Astronomy, astrophysics, aerospace engineering

BRAINSTORM: THREE TOPIC IDEAS I COULD SEE MYSELF ENJOYING

RED FLAGS TO AVOID
Some topics will hold you back before you begin. Steer clear of these:

- Topics that are too broad, such as “how to cure cancer”
- Topics with no measurable outcome
- Dangerous experiments
- Illegal activities
- Unethical research

2

Background Research

Learn what is already known before you experiment

Background research tells you what scientists have already discovered about your topic. It sharpens your question and points you toward a method. It also gives you the vocabulary that judges expect to hear.

WHERE TO RESEARCH

- Google Scholar (scholar.google.com)
- JSTOR, if your school has access
- Science journals such as Nature, Science, and PLOS
- Websites ending in .edu and .gov
- Library databases
- Science textbooks

WHAT TO RESEARCH

- History of the topic
- Current understanding
- Related experiments
- Key terms and definitions
- Possible experimental methods

HOW TO TAKE NOTES

- Use note cards or digital notes
- Record every source with its author, title, date, and web address
- Organize your notes by sub-topic

SAVE SOURCES AS YOU GO

Write down the full source details the moment you find something useful. Tracking citations during research saves hours of searching when you write your paper.

WORKSHEET • SOURCE LOG

Source (author, title, date)	What I learned

3

Formulating a Hypothesis

Turn your question into a testable prediction

A hypothesis is a testable prediction about what will happen in your experiment. The clearest hypotheses follow an if, then, because pattern that names what you will change, what you expect, and the reason behind it.

EXAMPLE HYPOTHESIS

If I increase the amount of fertilizer, **then** the plants will grow taller, **because** fertilizer provides nutrients that support growth.

Strong

A GOOD HYPOTHESIS IS

- Specific
- Testable
- Backed by a clear reason

Weak

A POOR HYPOTHESIS IS

- Vague
- Not testable
- Missing any reasoning

WORKSHEET • BUILD YOUR HYPOTHESIS

ATTEMPT 1

If _____
then _____
because _____

ATTEMPT 2 (REFINE YOUR WORDING)

4

Designing Your Experiment

Plan a fair test that changes one thing at a time

A well-designed experiment changes one factor while holding everything else steady. That is how you can trust that your results came from the thing you tested, and not from chance.

The three kinds of variables

Variable	Meaning
Independent	The one thing you change on purpose
Dependent	What you measure to see the effect
Controlled	Everything you keep the same

YOUR CONTROL GROUP

A control group is the version that receives no change. It gives you a baseline to compare against, so you can tell whether your independent variable made any difference at all.

SAMPLE SIZE

Run at least three to five trials. More trials make your results more reliable, because a single odd result has less power to mislead you.

EXPERIMENTAL DESIGN CHECKLIST

- Change only one variable
- Measure precisely
- Repeat your test several times
- Document everything

WORKSHEET • MY VARIABLES

INDEPENDENT VARIABLE (WHAT I WILL CHANGE)

DEPENDENT VARIABLE (WHAT I WILL MEASURE)

CONTROLLED VARIABLES (WHAT I WILL KEEP THE SAME)

5

Getting Approval

Some projects need permission before you begin

Certain projects must be reviewed and approved before any work starts. Check the rules for your competition early, because approval can take weeks. The categories below are the most common.

If your project involves...	You will need...
Human subjects	Institutional Review Board (IRB) approval and informed consent forms. Learn how to apply before you collect any data.
Vertebrate animals	Institutional Animal Care and Use Committee (IACUC) approval, humane treatment, and veterinary oversight.
Dangerous materials	Adult supervision and safety protocols for chemicals, biological agents, or radiation.

PLAN APPROVAL INTO YOUR TIMELINE

Never begin a regulated experiment before approval is granted. Doing so can disqualify an otherwise excellent project. Your fair or your teacher can point you to the right approval forms.

WORKSHEET • DOES MY PROJECT NEED APPROVAL?

- ☐ My project involves human subjects
- ☐ My project involves vertebrate animals
- ☐ My project involves chemicals, biological agents, or radiation
- ☐ None of the above apply

6

Conducting Your Experiment

Run your test and record everything carefully

This is where your plan becomes real data. Work carefully and stay safe. Write down what happens as it happens. Careful records are what separate a science project from a guess.

Lab notebook best practices

- Date every entry
- Write in pen, crossing out mistakes rather than erasing them
- Record everything, including attempts that did not work
- Include tables, photos, and sketches
- Have an adult witness sign your pages

SAFETY FIRST

- Wear a lab coat, goggles, and gloves when needed
- Keep an adult nearby for supervision
- Keep emergency contacts within reach
- Know where the fire extinguisher is

DATA COLLECTION

- Use tables and charts
- Stay consistent with your units
- Take photos and videos of the process

WORKSHEET • DATA TABLE

Trial	Measurement	Notes
1		
2		
3		
4		
5		

7

Analyzing Your Data

Turn your numbers into meaning

Analysis is where your raw measurements start to tell a story. Organize your data, picture it in a graph, and look honestly at what it shows about your hypothesis.

ORGANIZING DATA

- Build tables in Excel or Google Sheets
- Calculate averages and ranges

Choosing the right graph

Graph type	Use it when you are showing...
Bar graph	Comparisons across categories
Line graph	Change over time
Pie chart	Parts of a whole

Label your axes, give every graph a title, and use color with purpose rather than decoration.

FOR ADVANCED PROJECTS

- Mean, median, and mode
- Standard deviation
- T-tests
- Tools: Excel, Google Sheets, or R

INTERPRETING RESULTS

- Did the results support your hypothesis?
- Were there any unexpected findings?
- What sources of error existed?

WHAT PATTERNS DO I SEE IN MY DATA?

8

Drawing Conclusions

Answer your question with evidence

Your conclusion returns to the question you started with and answers it using your data. A strong conclusion is honest about what the evidence shows and what it does not.

A complete conclusion does the following:

- Answers your original question
- Explains how your data supports or refutes your hypothesis
- Discusses sources of error
- Connects your findings to the real world
- Suggests directions for future research

AN UNSUPPORTED HYPOTHESIS IS STILL A RESULT

If your data does not support your prediction, that is a real finding, not a failure. Explaining why is often the most interesting part of a project.

WORKSHEET • MY CONCLUSION

DID MY DATA SUPPORT MY HYPOTHESIS? HOW DO I KNOW?

WHAT COULD HAVE CAUSED ERROR, AND WHAT WOULD I DO DIFFERENTLY?

HOW COULD THIS MATTER IN THE REAL WORLD?

9

Creating Your Display Board

Tell your story at a glance

Your display board is the first thing a judge sees. A clear, well-organized board shows your work in a way that someone can follow in under a minute.

BOARD SPECIFICATIONS

- Tri-fold board, usually 36 inches by 48 inches
- Free-standing
- Check your competition size limits

DESIGN TIPS

- Use a readable font, at least 16 point for text and 24 point or larger for headings
- Leave white space so the board does not feel crowded
- Use color with restraint
- Print high-quality graphics
- Proofread for spelling and grammar
- Show photos of you doing the experiment

Suggested layout

Project Title · Name · School · Grade

LEFT PANEL

Question
Hypothesis
Materials
Procedure

CENTER PANEL

Data
Tables
Graphs
Photos of experiment

RIGHT PANEL

Results
Conclusions
Applications
Acknowledgments

KEEP THESE OFF YOUR BOARD

- Live or improperly preserved animals
- Chemicals
- Flames or fire
- Dangerous materials
- Unshielded batteries

The research paper is the written record of everything you did. It lets a reader repeat your work and understand your thinking from start to finish.

PAPER SECTIONS

- Title page
- Table of contents
- Abstract, a summary of about 250 words
- Introduction and background research
- Hypothesis
- Materials and methods
- Results
- Discussion and conclusion
- Bibliography
- Acknowledgments

FORMATTING

- Double-spaced
- 12 point Times New Roman
- One-inch margins
- Page numbers

CITATIONS

- Use MLA or APA format
- Cite all of your sources
- Include in-text citations and a bibliography

WORKSHEET • DRAFT PROGRESS

- ☐ Title page
- ☐ Abstract
- ☐ Introduction and background
- ☐ Hypothesis
- ☐ Materials and methods
- ☐ Results
- ☐ Discussion and conclusion
- ☐ Bibliography
- ☐ Acknowledgments

Judges want to meet the scientist behind the board. Your job is to explain your work clearly and show that you understand it deeply.

WHAT JUDGES WANT TO SEE

- Deep understanding of your project
- A clear explanation of your process
- Confident answers to their questions
- Real enthusiasm for your topic

PRESENTATION TIPS

- Make eye contact
- Speak clearly and at a steady pace
- Explain your board rather than reading it
- Show enthusiasm
- Be honest when you do not know an answer

PRACTICE TECHNIQUES

- Prepare a five-minute version of your project
- Practice with family and friends

QUESTIONS JUDGES OFTEN ASK

- Why did you choose this topic?
- What was your hypothesis?
- What were your results?
- What would you do differently?
- What are the real-world applications?

DRESS THE PART

Wear business casual clothing that is clean and neat, and choose comfortable shoes, since you will be on your feet for much of the day.

WORKSHEET • MY ANSWERS TO COMMON QUESTIONS

WHY DID I CHOOSE THIS TOPIC?

WHAT WOULD I DO DIFFERENTLY NEXT TIME?

Project Timeline Checklist

A science fair project rewards steady work over months. Use this timeline as a map. Adjust the dates to fit your own fair, then check off each milestone as you reach it.

When	Milestone	Target date	Done
6 months before	Choose your topic		☐
5 months before	Complete background research		☐
4 months before	Write your hypothesis and design your experiment		☐
3 months before	Get approvals, if your project needs them		☐
2 to 3 months before	Conduct your experiment		☐
1 month before	Analyze your data and draw conclusions		☐
3 weeks before	Create your display board		☐
2 weeks before	Write your research paper		☐
1 week before	Practice your presentation		☐
Day of	Arrive early and be confident		☐

START EARLIER THAN YOU THINK YOU NEED TO

The students who do best give themselves room for trials that fail and experiments that take longer than planned. Early starts turn into calm finishes.

Companion Resources

This guide pairs with a set of free downloads and video tutorials from the FairGame Foundation. Together they give you a template for every stage of your project.

Downloadable templates

Resource	Format
Complete Research Guide	PDF (this document)
Lab Notebook Template	PDF
Display Board Template	PowerPoint
Research Paper Template	Word
Project Timeline Checklist	Excel
Bibliography Template	Word
Hypothesis Worksheet	PDF
Data Collection Sheets	Excel

Video tutorials

- How to Choose a Great Project
- Designing a Controlled Experiment
- Creating Effective Data Visualizations
- Display Board Design Tips
- How to Present Like a Pro

Video tutorials are in production and will be released through the FairGame Foundation website.

You have everything you need. Now go build something you are proud of.

FairGame Foundation • Science Fairs Without Barriers